

Chap11-Exercises_LuisCorreia-745724_v1

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1 MAP5935 - Statistical Learning (Chapter 11 - Survival Analysis)

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<https://www.statlearning.com/>

```
[1]: import numpy as np
import random
import pandas as pd
import matplotlib.pyplot as plt
from matplotlib.pyplot import subplots

from ISLP import load_data
from ISLP.models import ModelSpec as MS

from sklearn.linear_model import \
    (LinearRegression ,
     LogisticRegression ,
     Lasso)
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import KFold
from sklearn.pipeline import Pipeline

from sklearn.model_selection import \
    (train_test_split ,
     GridSearchCV)

from lifelines import KaplanMeierFitter, CoxPHFitter
```

1.1 *Conceptual Exercises*

- 1.1.1 (8) Recall that the survival function $S(t)$, the hazard function $h(t)$, and the density function $f(t)$ are defined in equations (1 – 3) below, respectively. Furthermore, define $F(t) = 1 - S(t)$. Show that relationships A and B hold:

$$f(t) = dF(t)/dt \quad (A)$$

$$S(t) = \exp \left(- \int_0^t h(u) du \right) \quad (\text{B})$$

$$S(t) = \Pr(T > t) \quad (1)$$

$$h(t) = \lim_{\Delta t \rightarrow 0} \frac{\Pr(t < T \leq t + \Delta t | T > t)}{\Delta t} \quad (2)$$

$$f(t) = \lim_{\Delta t \rightarrow 0} \frac{\Pr(t < T \leq t + \Delta t)}{\Delta t} \quad (3)$$

Solution (A) Showing $f(t) = \frac{dF(t)}{dt}$

For any $\Delta t > 0$,

$$F(t) = 1 - S(t) = 1 - \Pr(T > t) = \Pr(T \leq t)$$

Then we have

$$F(t + \Delta t) - F(t) = \Pr(T \leq t + \Delta t) - \Pr(T \leq t) = \Pr(t < T \leq t + \Delta t).$$

Therefore, using the definition of $f(t)$ in equation (3), we have:

$$f(t) = \lim_{\Delta t \rightarrow 0} \frac{\Pr(t < T \leq t + \Delta t)}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{F(t + \Delta t) - F(t)}{\Delta t} = \frac{dF(t)}{dt},$$

i.e. f is the derivative of the cdf F .

Solution (B) Expressing $S(t)$ in terms of the hazard $h(t)$

First note that, using the definition of conditional probability,

$$\Pr(t < T \leq t + \Delta t | T > t) = \frac{\Pr(t < T \leq t + \Delta t, T > t)}{\Pr(T > t)} = \frac{\Pr(t < T \leq t + \Delta t)}{S(t)}.$$

Plugging this into the definition of the hazard,

$$h(t) = \lim_{\Delta t \rightarrow 0} \frac{\Pr(t < T \leq t + \Delta t | T > t)}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{\Pr(t < T \leq t + \Delta t)}{\Delta t S(t)} = \lim_{\Delta t \rightarrow 0} \frac{\Pr(t < T \leq t + \Delta t)}{\Delta t} \frac{1}{S(t)} = \frac{f(t)}{S(t)}.$$

Thus

$$f(t) = h(t) S(t). \quad (1)$$

We also have $F(t) = 1 - S(t)$ and (1), so

$$\frac{dS(t)}{dt} = -\frac{dF(t)}{dt} = -f(t) = -h(t) S(t).$$

Hence S satisfies the differential equation

$$\frac{dS(t)}{dt} = -h(t) S(t), \quad S(0) = \Pr(T > 0) = 1.$$

Solving it by substitution of variables:

$$\frac{dS(t)}{S(t)} = -h(t) dt \implies \int_{S(0)}^{S(t)} \frac{1}{s} ds = -\int_0^t h(u) du,$$

so

$$\log S(t) - \log S(0) = -\int_0^t h(u) du.$$

Since $S(0) = 1$, $\log S(0) = 0$, and we obtain

$$\log S(t) = -\int_0^t h(u) du \implies S(t) = \exp\left(-\int_0^t h(u) du\right).$$

1.2 *Applied Exercises*

1.2.1 (10) This exercise focuses on the brain tumor data, which is included in the ISLP library.

1.2.2 (a) Plot the Kaplan-Meier survival curve with ± 1 standard error bands, using the `KaplanMeierFitter()` estimator in the `lifelines` package.

A `BrainCancer` data set consisting of survival times for patients diagnosed with brain cancer.

- **sex:** Factor with levels “Female” and “Male”
- **diagnosis:** Factor with levels “Meningioma”, “LG glioma”, “HG glioma”, and “Other”.
- **loc:** Location factor with levels “Infratentorial” and “Supratentorial”.
- **ki:** Karnofsky index.

The **Karnofsky Performance Status (KPS)** scale is a standard way of measuring a patient’s general well-being and ability to perform ordinary daily tasks. It is widely used by healthcare professionals, particularly in oncology and palliative care, to assess a patient’s functional impairment, determine their prognosis, and help guide treatment decisions.

- **gtv:** Gross tumor volume, in cubic centimeters.

- **stereo**: Stereotactic method factor with levels “SRS” and “SRT”.

The **primary difference** between *stereotactic radiosurgery (SRS)* and *stereotactic radiotherapy (SRT)* is the **number of treatment sessions (fractions)**. SRS delivers a single, high dose of radiation in one session, while SRT divides the total radiation dose into multiple smaller doses administered over several days or weeks

- **status**: Whether the patient is still alive at the end of the study: 0=Yes, 1=No.
- **time**: Survival time, in months

```
[2]: # Load data
BrainCancer = load_data("BrainCancer")
BrainCancer.head()
```

```
[2]:      sex  diagnosis      loc  ki   gtv stereo  status   time
0  Female  Meningioma  Infratentorial  90   6.11    SRS      0  57.64
1   Male   HG glioma  Supratentorial  90  19.35    SRT      1   8.98
2  Female  Meningioma  Infratentorial  70   7.95    SRS      0  26.46
3  Female   LG glioma  Supratentorial  80   7.61    SRT      1  47.80
4   Male   HG glioma  Supratentorial  90   5.06    SRT      1   6.30
```

```
[3]: durations = BrainCancer["time"].values
events = BrainCancer["status"].values    # 1 = death, 0 = censored

# Fit Kaplan-Meier estimator
kmf = KaplanMeierFitter()
kmf.fit(durations=durations, event_observed=events, label="Kaplan-Meier")

# --- Greenwood standard errors -----
# event_table has index = times, columns: 'at_risk', 'observed', 'censored', ...
ev_tab = kmf.event_table

n = ev_tab["at_risk"].values          # n_i
d = ev_tab["observed"].values         # d_i
S = kmf.survival_function_["Kaplan-Meier"].values # \hat{S}(t_j)

# Greenwood sum: cumulative sum of d_i / [n_i (n_i - d_i)]
term = np.zeros_like(d, dtype=float)
mask = (n > 0) & (n - d > 0)          # avoid division by zero at the very end
term[mask] = d[mask] / (n[mask] * (n[mask] - d[mask]))

greenwood_sum = np.cumsum(term)
var_S = (S ** 2) * greenwood_sum
se_S = np.sqrt(var_S)

# Construct ± 1 SE bands, truncated to [0, 1]
lower = np.clip(S - se_S, 0, 1)
upper = np.clip(S + se_S, 0, 1)
```

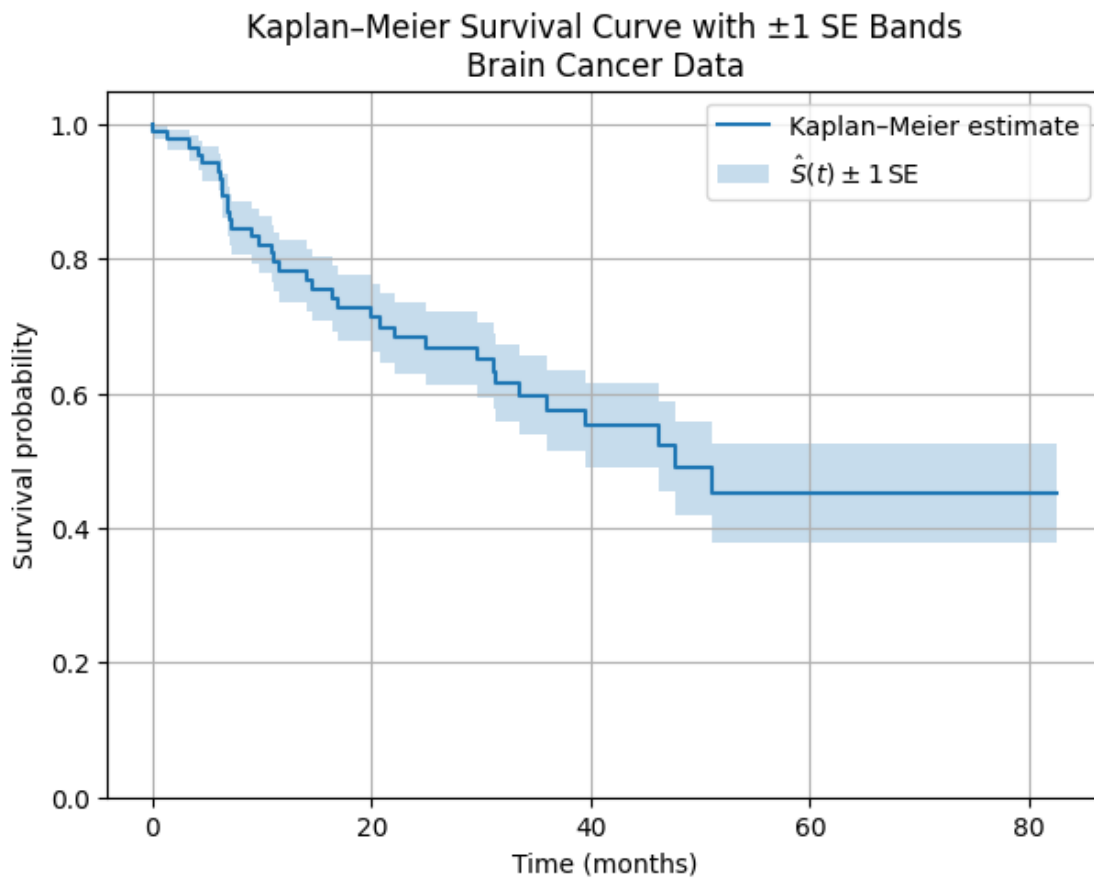
```

t = kmf.survival_function_.index

# --- Plot KM curve with ±1 SE bands -----
plt.figure(figsize=(7, 5))
plt.step(t, S, where="post", label="Kaplan-Meier estimate")
plt.fill_between(t, lower, upper, step="post", alpha=0.25,
                 label=r"$\hat{S}(t) \pm 1\, \mathrm{SE}$")

plt.xlabel("Time (months)")
plt.ylabel("Survival probability")
plt.ylim(0, 1.05)
plt.title("Kaplan-Meier Survival Curve with ±1 SE Bands\nBrain Cancer Data")
plt.legend()
plt.grid(True)
plt.show()

```



Comments

- The Kaplan–Meier curve shows the estimated probability of remaining alive as a function

of time since diagnosis. The stepwise blue line starts at 1 (all patients alive at time 0) and decreases over time as deaths occur, reaching about 0.45 by 80 months, which means that roughly 45% of the cohort are still alive at that time.

- The shaded band represents $\hat{S}(t) \pm 1$ standard error, so it gives a sense of the statistical uncertainty around the estimated survival curve: it is narrow at early times, when many patients are still under observation, and becomes wider at later times, where there are fewer patients at risk and the survival estimates are less precise.
- Overall, the plot suggests a **gradual decline in survival with increasing uncertainty toward the end of follow-up**.

1.2.3 (b) Draw a bootstrap sample of size $n = 88$ from the pairs (y_i, δ_i) , and compute the resulting Kaplan-Meier survival curve. Repeat this process $B = 200$ times. Use the results to obtain an estimate of the standard error of the Kaplan-Meier survival curve at each timepoint. Compare this to the standard errors obtained in (a).

```
[4]: # -----
# Original KM fit and Greenwood standard errors (from part a)
# -----
durations = BrainCancer["time"].values
events = BrainCancer["status"].values    # 1 = death, 0 = censored
n = len(BrainCancer)                    # should be 88

kmf = KaplanMeierFitter()
kmf.fit(durations, events)

# Time grid and KM estimate
t_grid = kmf.survival_function_.index.values
S_hat = kmf.survival_function_.iloc[:, 0].values

# Greenwood SE
ev_tab = kmf.event_table
n_at_risk = ev_tab["at_risk"].values
d_obs = ev_tab["observed"].values

term = np.zeros_like(d_obs, dtype=float)
mask = (n_at_risk > 0) & (n_at_risk - d_obs > 0)
term[mask] = d_obs[mask] / (n_at_risk[mask] * (n_at_risk[mask] - d_obs[mask]))

greenwood_sum = np.cumsum(term)
var_greenwood = (S_hat ** 2) * greenwood_sum
se_greenwood = np.sqrt(var_greenwood)

# -----
# Bootstrap estimate of the standard error of S(t)
# -----
B = 200
```

```

boot_surv = np.empty((B, len(t_grid)))

rng = np.random.default_rng(seed=123)    # fix seed for reproducibility

for b in range(B):
    # sample n pairs (time, status) with replacement
    idx = rng.integers(low=0, high=n, size=n)
    durations_b = durations[idx]
    events_b = events[idx]

    kmf_b = KaplanMeierFitter()
    kmf_b.fit(durations_b, events_b)

    # Evaluate the bootstrap KM curve on the original time grid
    boot_surv[b, :] = kmf_b.predict(t_grid)

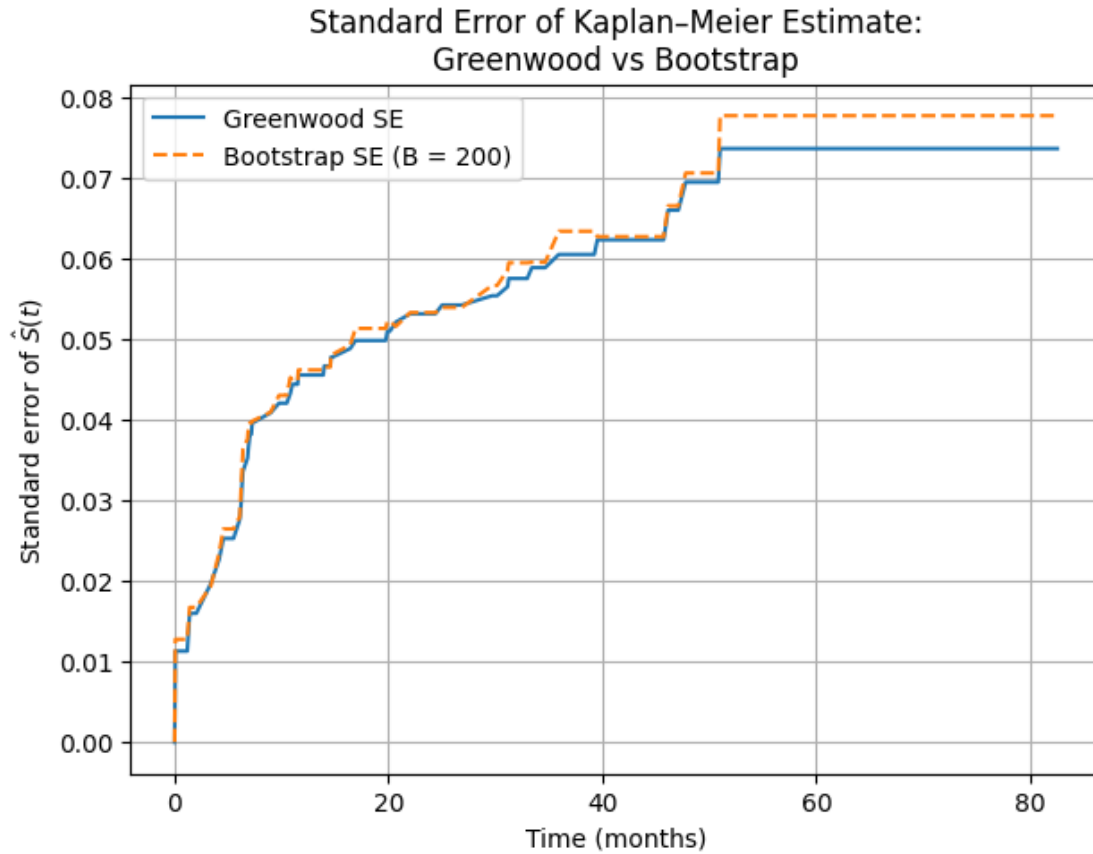
# Bootstrap SE at each time point:
#  $\widehat{\mathrm{SE}}_{\text{boot}}(\hat{S}(t_j))$ 
se_boot = boot_surv.std(axis=0, ddof=1)

# -----
# Compare Greenwood vs bootstrap SEs
# -----

plt.figure(figsize=(7, 5))
plt.plot(t_grid, se_greenwood, label="Greenwood SE", linestyle="-")
plt.plot(t_grid, se_boot, label="Bootstrap SE (B = 200)", linestyle="--")

plt.xlabel("Time (months)")
plt.ylabel(r"Standard error of  $\hat{S}(t)$ ")
plt.title("Standard Error of Kaplan-Meier Estimate:\nGreenwood vs Bootstrap")
plt.legend()
plt.grid(True)
plt.show()

```



Comments The *Greenwood* and *Bootstrap* standard errors are very similar across the follow-up period, with only small differences in the right tail where the bootstrap SE is slightly larger, suggesting that **Greenwood's formula may mildly underestimate uncertainty at the latest times.**

1.2.4 (c) Fit a Cox proportional hazards model that uses all of the predictors to predict survival. Summarize the main findings.

```
[5]: # Create design matrix: all predictors, with dummy variables for categorical
X = BrainCancer.drop(columns=["time", "status"])
X = pd.get_dummies(X, drop_first=True)

# Combine with survival outcome
df_cox = pd.concat([BrainCancer[["time", "status"]], X], axis=1)

# Fit Cox proportional hazards model
cph = CoxPHFitter()
cph.fit(df_cox, duration_col="time", event_col="status")
```

```
# Show summary (coefficients, SE, z, p, etc.)
#cph.print_summary()
cph.summary
```

```
[5]:
```

	coef	exp(coef)	se(coef)	coef lower 95%	\
covariate					
ki	-0.057920	0.943725	0.018313	-0.093814	
gtv	0.036041	1.036699	0.022623	-0.008298	
sex_Male	0.097854	1.102802	0.360701	-0.609107	
diagnosis_LG glioma	-1.091266	0.335791	0.573943	-2.216172	
diagnosis_Meningioma	-2.082012	0.124679	0.453688	-2.971225	
diagnosis_Other	-1.235849	0.290588	0.625866	-2.462524	
loc_Supratentorial	0.375835	1.456207	0.697787	-0.991803	
stereo_SRT	0.139636	1.149855	0.598031	-1.032483	

	coef upper 95%	exp(coef) lower 95%	\
covariate			
ki	-0.022026	0.910453	
gtv	0.080381	0.991736	
sex_Male	0.804815	0.543836	
diagnosis_LG glioma	0.033641	0.109026	
diagnosis_Meningioma	-1.192799	0.051240	
diagnosis_Other	-0.009174	0.085220	
loc_Supratentorial	1.743473	0.370907	
stereo_SRT	1.311754	0.356122	

	exp(coef) upper 95%	cmp to	z	p	\
covariate					
ki	0.978214	0.0	-3.162720	0.001563	
gtv	1.083699	0.0	1.593157	0.111125	
sex_Male	2.236283	0.0	0.271288	0.786169	
diagnosis_LG glioma	1.034213	0.0	-1.901350	0.057256	
diagnosis_Meningioma	0.303371	0.0	-4.589080	0.000004	
diagnosis_Other	0.990868	0.0	-1.974623	0.048311	
loc_Supratentorial	5.717165	0.0	0.538610	0.590156	
stereo_SRT	3.712681	0.0	0.233493	0.815379	

	-log2(p)
covariate	
ki	9.321444
gtv	3.169745
sex_Male	0.347088
diagnosis_LG glioma	4.126426
diagnosis_Meningioma	17.777101
diagnosis_Other	4.371505
loc_Supratentorial	0.760832
stereo_SRT	0.294458

Solution : We fitted a Cox proportional hazards model with all predictors (`sex`, `diagnosis`, `loc`, `ki`, `gtv`, `stereo`) to the brain-cancer survival data (`time`, `status`). Below we summarize the main findings based on the entries in `cph.summary` (coefficients, hazard ratios, confidence intervals, and p -values).

- The continuous covariate **ki** has a negative coefficient ($\hat{\beta} = -0.06$, $p < 0.005$), with hazard ratio $\exp(\hat{\beta}) = 0.94$ (95% CI: 0.91–0.98).
Thus, **larger values of ki are associated with a lower hazard of death**: every one-unit increase in **ki** multiplies the hazard by about 0.94.
- The categorical variable **diagnosis** is also strongly related to survival. The reference category is **high-grade glioma (HG glioma)**, so the three diagnosis coefficients represent the effect relative to HG glioma:
 - **Meningioma** has $\hat{\beta} = -2.08$, $\exp(\hat{\beta}) = 0.12$ (95% CI: 0.05–0.30, $p < 0.005$), indicating that the hazard for these patients is about **88% lower** than for HG glioma.
 - **Other** diagnoses have $\exp(\hat{\beta}) = 0.29$ (95% CI: 0.09–0.99, $p \approx 0.05$), also suggesting a substantially lower hazard than HG glioma.
 - **Low-grade glioma (LG glioma)** has $\exp(\hat{\beta}) = 0.34$ (95% CI: 0.11–1.03, $p \approx 0.06$), pointing to a reduced hazard with borderline statistical significance.

Overall, **patients with HG glioma have the poorest prognosis**, while those with meningioma (and, to a lesser extent, “other” and LG glioma) have better survival.

- The remaining covariates show no clear evidence of association with hazard after adjusting for **ki** and **diagnosis**:
 - **Tumor volume gtv** has $\exp(\hat{\beta}) = 1.04$ (95% CI: 0.99–1.08, $p = 0.11$), suggesting at most a weak positive effect.
 - **Sex (Male vs Female)**, **tumor location** (supratentorial vs infratentorial), and **stereotactic method** (SRT vs SRS) all have hazard ratios close to 1, with wide confidence intervals including 1 and large p -values ($p > 0.5$), so **there is no strong evidence that these factors affect survival once the other predictors are in the model**.

In summary, the Cox model fitted to this dataset suggests that survival is mainly driven by **diagnosis** and **ki**, with high-grade glioma and smaller values of **ki** associated with markedly worse outcomes, whereas sex, tumor location, stereotactic method, and tumor volume appear to play only a minor role after adjustment.

1.2.5 (d) Stratify the data by the value of ki. (Since only one observation has **ki**= 40, you can group that observation together with the observations that have **ki**= 60.) Plot Kaplan-Meier survival curves for each of the five strata, adjusted for the other predictors.

```
[6]: # -----
# 1. Load data and create Karnofsky strata (5 levels)
#   Group ki == 40 together with ki == 60
# -----

BrainCancer["ki_group"] = BrainCancer["ki"].replace({40: 60})
```

```

ki_levels = np.sort(BrainCancer["ki_group"].unique()) # e.g. [60, 70, 80, 90, 100]

# -----
# 2. Fit a Cox model with all predictors, using numeric/dummy
#     covariates (this will be used to get adjusted survival curves)
# -----
# Keep original predictors, but drop the numeric ki (we now use ki_group)
predictors = ["sex", "diagnosis", "loc", "stereo", "gtv", "ki_group"]

df = BrainCancer[predictors + ["time", "status"]].copy()
df["ki_group"] = df["ki_group"].astype("category")

# Dummy-code categorical predictors (including ki_group)
X = pd.get_dummies(df.drop(columns=["time", "status"]), drop_first=True)
df_cox = pd.concat([df[["time", "status"]], X], axis=1)

cph = CoxPHFitter()
cph.fit(df_cox, duration_col="time", event_col="status")

# -----
# 3. Build representative covariate profiles:
#     - For each ki stratum, set ki_group to that value
#     - Other predictors set to mean (numeric) or mode (categorical)
# -----
rep = {
    "gtv": BrainCancer["gtv"].mean(),
    "sex": BrainCancer["sex"].mode()[0],
    "diagnosis": BrainCancer["diagnosis"].mode()[0],
    "loc": BrainCancer["loc"].mode()[0],
    "stereo": BrainCancer["stereo"].mode()[0],
}

profiles = []
index_names = []

for k in ki_levels:
    row = rep.copy()
    row["ki_group"] = k
    profiles.append(row)
    index_names.append(f"ki = {k}")

profiles = pd.DataFrame(profiles)
profiles["ki_group"] = profiles["ki_group"].astype("category")
profiles.index = index_names

# Dummy-code profiles using the SAME encoding as the Cox model

```

```

profiles_dummies = pd.get_dummies(profiles, drop_first=True)
profiles_dummies = profiles_dummies.reindex(
    columns=df_cox.drop(columns=["time", "status"]).columns,
    fill_value=0,
)

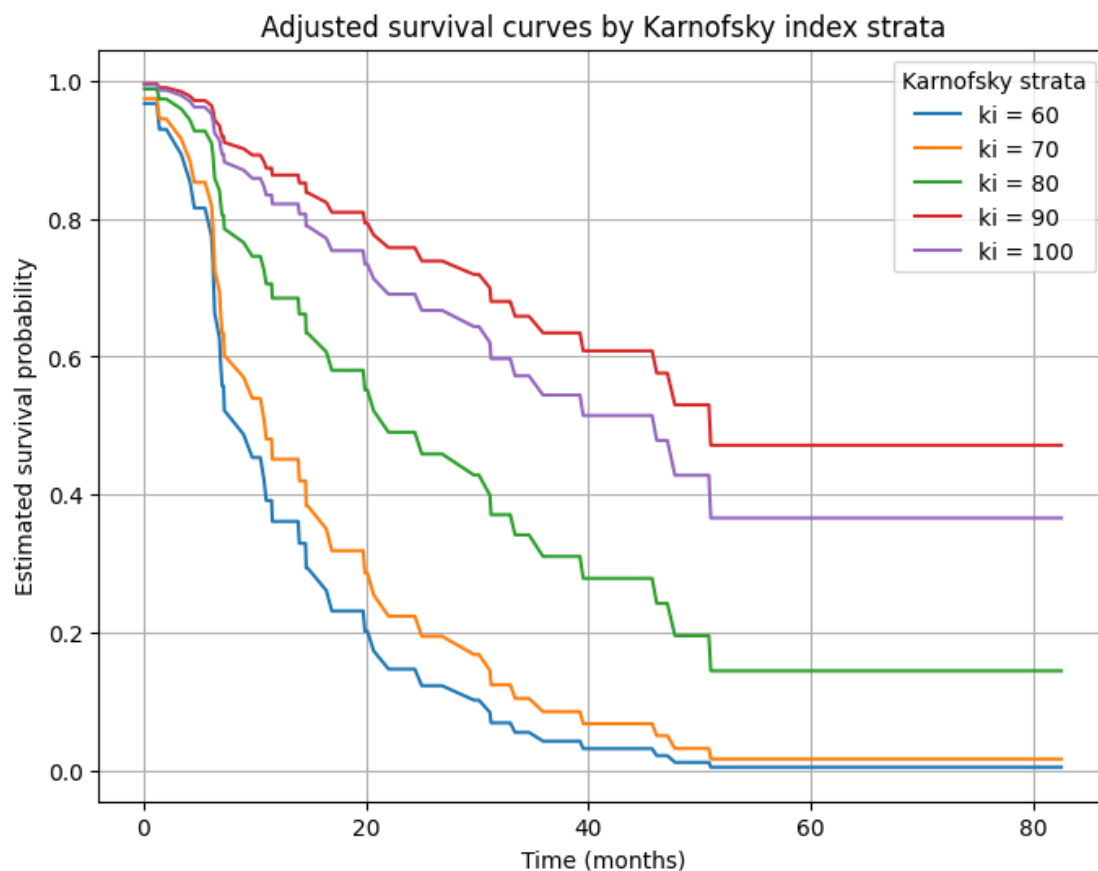
# -----
# 4. Predict survival curves for each ki stratum (adjusted for
#     the other predictors) and plot them
# -----
surv_curves = cph.predict_survival_function(profiles_dummies)

fig, ax = plt.subplots(figsize=(8, 6))
surv_curves.plot(ax=ax)

ax.set_xlabel("Time (months)")
ax.set_ylabel("Estimated survival probability")
ax.set_title("Adjusted survival curves by Karnofsky index strata")
ax.legend(title="Karnofsky strata")
ax.grid(True)

plt.show()

```



Comments The adjusted survival curves by ki show a clear and monotonic pattern: **patients with higher Karnofsky scores have consistently better survival**. - The curves for ki = 90 and ki = 100 stay highest over the whole follow-up, with more than half of those patients still alive after about 40–50 months. - The curves for ki = 60 and ki = 70 drop rapidly, approaching zero survival by around 40 months. - The ki = 80 group lies in between. - Since these curves are adjusted for diagnosis, tumor volume, sex, location, and stereotactic method via the Cox model, this pattern suggests that **baseline functional status (ki) is an important independent predictor of survival**: the better the performance score, the lower the estimated hazard and the higher the probability of long-term survival.